CHAPTER 5

Eigenvalues and Eigenvectors

特征值与特征向量

Chapter 5 Eigenvalues and Eigenvectors

- § 5.1 Eigenvectors and Eigenvalues
- § 5.2 The Characteristic Equation
- § 5.3 Diagonalization
- § 5.4 Eigenvectors and Linear Transformations
- § 5.5 Complex Eigenvalues
- § 5.6 Discrete Dynamical Systems

EXAMPLE: Let $A = \begin{bmatrix} 0 & -2 \\ -4 & 2 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Examine the images of **u** and **v** under multiplication by A.

Solution

$$A\mathbf{u} = \begin{bmatrix} 0 & -2 \\ -4 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix} = -2\begin{bmatrix} 1 \\ 1 \end{bmatrix} = -2\mathbf{u}$$

$$A\mathbf{v} = \begin{bmatrix} 0 & -2 \\ -4 & 2 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix} \neq \lambda\mathbf{v}$$

u is called an *eigenvector* of A.

v is not an eigenvector of A since $A\mathbf{v}$ is not a multiple of **v**.

§ 5.1 Eigenvectors and Eigenvalues

DEFINITION

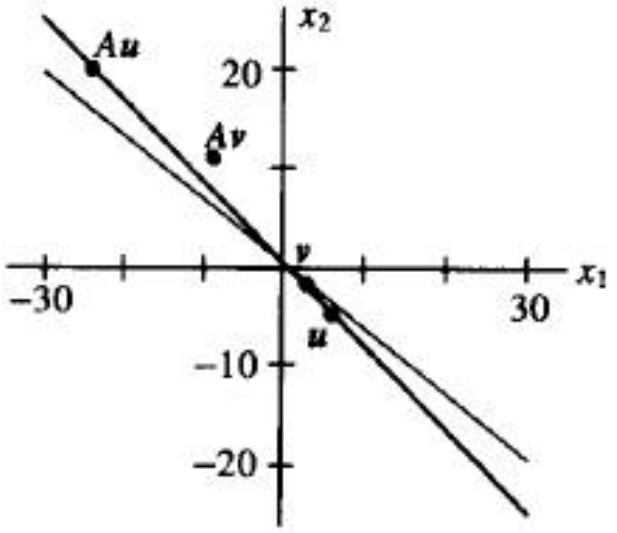
An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$; such an $\mathbf{x}$ is called an *eigenvector corresponding to* λ.

EXAMPLE: Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$.

Examine the images of $\mathbf{u}$ and $\mathbf{v}$ under multiplication by A.

$$Au = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix} = -4u$$

$$Av = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -9 \\ 11 \end{bmatrix} \neq \lambda \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$



$Au=-4u$，但 $Av\neq\lambda v$

- u is an eigenvector corresponding to an eigenvalue (-4)
- v is not an eigenvector of A

EXAMPLE: Show that 4 is an eigenvalue of $A = \begin{bmatrix} 0 & -2 \\ -4 & 2 \end{bmatrix}$ and find the corresponding eigenvectors.

Solution: Scalar 4 is an eigenvalue of A if and only if $A\mathbf{x} = 4\mathbf{x}$ has a nontrivial solution.

$$A\mathbf{x} - 4\mathbf{x} = \mathbf{0}$$

$$(A - 4I)\mathbf{x} = \mathbf{0}.$$

$$A - 4I = \begin{bmatrix} 0 & -2 \\ -4 & 2 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} -4 & -2 \\ -4 & -2 \end{bmatrix}$$

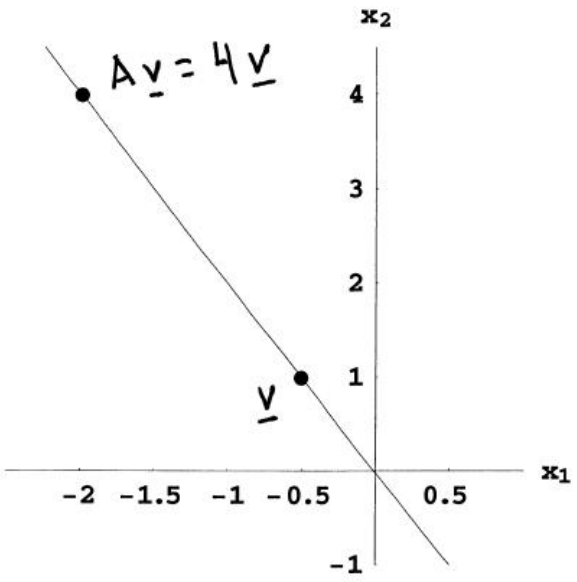
Now solve $(A-4I)\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} -4 & -2 & 0 \\ -4 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \qquad \mathbf{x} = \begin{bmatrix} \frac{-1}{2}x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} \frac{-1}{2} \\ 1 \end{bmatrix}.$$

Each vector of the form $x_2 \begin{bmatrix} \frac{-1}{2} \\ 1 \end{bmatrix}$ is an eigenvector corresponding to the eigenvalue $\lambda = 4$.

§ 5.1 Eigenvectors and Eigenvalues



Eigenspace for $\lambda = 4$

The set of all solutions to $(A-\lambda I)\mathbf{x} = \mathbf{0}$ is called the **eigenspace** of A corresponding to λ.

Warning: The method just used to find eigen*vectors*, ***cannot*** be used to find eigen*values*.

How do we find the eigenvalues λ?

x must be nonzero

$\Downarrow$

$(A - \lambda I)\mathbf{x} = \mathbf{0}$ must have nontrivial solutions

$\Downarrow$

$(A - \lambda I)$ is not invertible

$\Downarrow$

$$\det(A - \lambda I) = 0$$

(called the *characteristic equation*)

Solve $\det(A - \lambda I) = 0$ for λ to find the eigenvalues.

Characteristic polynomial: $\det(A - \lambda I)$

Characteristic equation: $\det(A - \lambda I) = 0$

EXAMPLE: Find the eigenvalues of $A = \begin{bmatrix} 0 & 1 \\ -6 & 5 \end{bmatrix}$.

Solution: Since

$$A - \lambda I = \begin{bmatrix} 0 & 1 \\ -6 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -\lambda & 1 \\ -6 & 5-\lambda \end{bmatrix},$$

the equation $\det(A - \lambda I) = 0$ becomes

$$-\lambda(5-\lambda) + 6 = 0$$

$$\lambda^2 - 5\lambda + 6 = 0$$

Factor:

$$(\lambda - 2)(\lambda - 3) = 0.$$

So the eigenvalues are 2 and 3.

For a 3×3 matrix or larger, recall that a determinant can be computed by cofactor expansion.

§ 5.1 Eigenvectors and Eigenvalues

THEOREM 1 The eigenvalues of a triangular matrix are the entries on its main diagonal.

Proof for the 3×3 *Upper Triangular Case*: Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}.$$

and then

$$A - \lambda I = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ 0 & a_{22} - \lambda & a_{23} \\ 0 & 0 & a_{33} - \lambda \end{bmatrix}.$$

By definition, λ is an eigenvalue of A if and only if $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution. This occurs if and only if $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a free variable.

When does this occur?

$$a_{11} - a_{11} - \lambda = 0 \Rightarrow \lambda = a_{11}$$

$$a_{22} - a_{22} - \lambda = 0 \Rightarrow \lambda = a_{22}$$ 22

$$a_{33} - a_{33} - \lambda = 0 \Rightarrow \lambda = a_{33}$$ 33

- Example： Let $A=\begin{bmatrix}3 & 6 & -8\\ 0 & 0 & 6\\ 0 & 0 & 2\end{bmatrix}, B=\begin{bmatrix}4 & 0 & 0\\ -2 & 1 & 0\\ 5 & 3 & 4\end{bmatrix}$,

By Theorem 1,

The eigenvalues of A are 3, 0 an 2.

The eigenvalues of B are 4 and 1.

What does it mean for a matrix A to have a eigenvalue of 0?

- A matrix A has an eigenvalue of 0, if and only if the equation

$$Ax = 0x$$

has a nontrivial solution. It means A is not invertible.

So the Invertible Matrix Theorem can be continued:

0 is an eigenvalue of A if and only if A is not invertible.

§ 5.1 Eigenvectors and Eigenvalues

THEOREM 2 If $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \ldots, \lambda_r$ of an $n \times n$ matrix A, then $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is a linearly independent set.

Proof： Suppose $\{v_1,\ldots,v_r\}$ is linearly dependent. Then there exist scalar $c_1,\ldots,c_p$ such that

$$c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{v}_{p+1} \qquad (1)$$

and $\{v_1, v_2, \ldots, v_p\}$ is linearly independent. Multiplying both sides by A and using $Av_k = \lambda_k v_k$ for each k

$$c_1A\mathbf{v}_1 + \cdots + c_pA\mathbf{v}_p = A\mathbf{v}_{p+1}$$
$$c_1\lambda_1\mathbf{v}_1 + \cdots + c_p\lambda_p\mathbf{v}_p = \lambda_{p+1}\mathbf{v}_{p+1} \qquad (2)$$

(1) $\times$ $\boldsymbol{\lambda}_{p+1}$ - (2)

$$c_1(\lambda_1 - \lambda_{p+1})\mathbf{v}_1 + \cdots + c_p(\lambda_p - \lambda_{p+1})\mathbf{v}_p = \mathbf{0} \quad (3)$$

Since $\{v_1,\ldots,v_p\}$ is linearly independent, the weights in (3) are all zero.

But the factors $\boldsymbol{\lambda}_i$- $\boldsymbol{\lambda}_{p+1}$ are nonzero, because the eignvalues are distinct. Hence $c_i = 0$. It is opposite to the (1)

So $\{v_1,\ldots,v_r\}$ is linearly independent.

Assignment

- Ex5.1: 23,25

思考题

Problem:

Suppose λ is an eigenvalue of A.

What is the eigenvalue of A^n ?

不难证明：若λ是A的特征值，则λ^k 是A^k的特征值；$\varphi(\lambda)$ 是$\varphi(A)$的特征值（ 其中$\varphi(\lambda) = a_0 + a_1\lambda + \ldots + a_m\lambda^m$是$\lambda$的多项式，$\varphi(A) = a_0E + a_1A + \ldots + a_mA^m$是矩阵$A$的多项式 ）

§ 5.2 The Characteristic Equation

- 1. Determinants
- 2. The Characteristic Equation
- 3. Similarity

§ 5.2 The Characteristic Equation

Before turning to larger matrices, we summarize the properties of determinants needed to study eigenvalues.

1. Determinants（回顾相关属性）

Suppose the echelon form U is obtained from A by a sequence of row replacements or interchanges, but without scaling.

$$A \sim U = \begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots & u_{1n} \\ 0 & u_{22} & u_{23} & \cdots & u_{2n} \\ 0 & 0 & u_{33} & \cdots & u_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & u_{nn} \end{bmatrix}$$

§ 5.2 The Characteristic Equation

The **determinant** of A, written $\det A$, is defined as follows:

$$\det A = \begin{cases} (-1)^r \cdot \begin{pmatrix} \text{product of} \\ \text{pivots in } U \end{pmatrix}, & \text{when } A \text{ is invertible} \\ 0, & \text{when } A \text{ is not invertible} \end{cases}$$

(r is the number of row interchanges)

§ 5.2 The Characteristic Equation

The formula for the determinant shows that A is invertible if and only if det A is nonzero.

It can be added to the Invertible Matrix Theorem.

THEOREM (The Invertible Matrix Theorem - continued)

Let A be an $n \times n$ matrix. Then A is invertible if and only if:

s. The number 0 is not an eigenvalue of A.

t. $\det A \neq 0$

§ 5.2 The Characteristic Equation

- **Theorem 3 Properties of Determinants**

  Let A and B be n × n matrices.

  a. A is invertible if and only if det A ≠ 0.

  b. det AB = (det A)(det B).

  c. det A^T = det A.

  d. If A is triangular, then det A is the product of the entries on the main diagonal of A.

§ 5.2 The Characteristic Equation

- **2. The Characteristic Equation**

The **scalar** equation det (A - λI) = 0 is called the characteristic equation of A.

A scalar λ is an eigenvalue of an n×n matrix A if and only if λ satisfies the characteristic equation

$$\det(A-\lambda I)=0$$

EXAMPLE: Find the eigenvalues of $A = \begin{bmatrix} 3 & 2 & 3 \\ 0 & 6 & 10 \\ 0 & 0 & 2 \end{bmatrix}$.

Solution:

$$\det(A - \lambda I) = \det\begin{bmatrix} 3-\lambda & 2 & 3 \\ 0 & 6-\lambda & 10 \\ 0 & 0 & 2-\lambda \end{bmatrix}$$

Characteristic equation:

$$(3-\lambda)(6-\lambda)(2-\lambda) = 0.$$

eigenvalues: 3, 6, 2

EXAMPLE: Find the eigenvalues of $A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Solution:

$$\det(A - \lambda I) = \det\begin{bmatrix} 5-\lambda & -2 & 6 & -1 \\ 0 & 3-\lambda & -8 & 0 \\ 0 & 0 & 5-\lambda & 4 \\ 0 & 0 & 0 & 1-\lambda \end{bmatrix}$$

$$= (5-\lambda)(3-\lambda)(5-\lambda)(1-\lambda)$$

Characteristic equation:

$$(5-\lambda)^2(3-\lambda)(1-\lambda) = 0$$

eigenvalues: 3, 5, 1

Some Concept:

- 1. If A is an n×n matrix, then det (A- λI)is a polynomial of degree n called the characteristic polynomial of A.
- 2. The (**algebraic**) **multiplicity** of an eigenvalue is its multiplicity as a root of the characteristic equation.

eg. In the upper example, the eigenvalue 5 has *multiplicity* 2 because (λ-5) occurs two times as a factor of the characteristic polynomial.

EXAMPLE: Find the characteristic polynomial of

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 5 & 3 & 0 & 0 \\ 9 & 1 & 3 & 0 \\ 1 & 2 & 5 & -1 \end{bmatrix}$$

and then find all the eigenvalues and the algebraic multiplicity of each eigenvalue.

Solution:

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 0 & 0 & 0 \\ 5 & 3-\lambda & 0 & 0 \\ 9 & 1 & 3-\lambda & 0 \\ 1 & 2 & 5 & -1-\lambda \end{vmatrix}$$

$$= (2-\lambda)(3-\lambda)(3-\lambda)(-1-\lambda) = 0$$

eigenvalues: 2, 3, −1

- Example： The characteristic polynomial of a 6×6 matrix is

$$\lambda^6 - 4\lambda^5 - 12\lambda^4$$

Find the eigenvalues and their multiplicities.

Solution：

$$\lambda^6 - 4\lambda^5 - 12\lambda^4 = \lambda^4(\lambda^2 - 4\lambda - 12) = \lambda^4(\lambda - 6)(\lambda + 2)$$

The eigenvalues are

0 (multiplicity 4), 6 (multiplicity 1), -2 (multiplicity 1).

- **3. Similarity（ 相似 ）**

For $n \times n$ matrices A and B, we say the A is **similar** to B if there is an invertible matrix P such that

$$P^{-1}AP = B \qquad \text{or equivalently,} \qquad A = PBP^{-1}.$$

§ 5.2 The Characteristic Equation

Theorem 4: If $n \times n$ matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

Proof: If $B = P^{-1}AP$, then

$$B - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}(AP - \lambda P) = P^{-1}(A - \lambda I)P$$

$$\begin{aligned}\det(B - \lambda I) &= \det[P^{-1}(A - \lambda I)P] \\ &= \det(P^{-1}) \cdot \det(A - \lambda I) \cdot \det(P)\end{aligned}$$

Because $\det(P^{-1}) \cdot \det(P) = \det(P^{-1}P) = \det I = 1$,

So we have $\det(B - \lambda I) = \det(A - \lambda I)$

§ 5.2 The Characteristic Equation

- **4. Application to Dynamical Systems**

Example：

Let $A = \begin{bmatrix} .95 & .03 \\ .05 & .97 \end{bmatrix}$. Analyze the long-term behavior of a dynamical system by $x_{k+1} = Ax_k$ (k = 0,1,2,…) with $x_0 = \begin{bmatrix} .6 \\ .4 \end{bmatrix}$

Sol.

$$0 = \det \begin{bmatrix} 0.95-\lambda & 0.03 \\ 0.05 & 0.97-\lambda \end{bmatrix} = (0.95-\lambda)(0.97-\lambda) - (0.03)(0.05)$$

$$= \lambda^2 - 1.92\lambda + 0.92$$

$$\lambda = \frac{1.92 \pm \sqrt{(1.92)^2 - 4(0.92)}}{2} = \frac{1.92 \pm \sqrt{0.0064}}{2}$$

$$= \frac{1.92 \pm 0.08}{2} = 1 \text{ 或 } 0.92$$

It is readily checked that eigenvectors corresponding to $\lambda=1$ and $\lambda=0.92$ are multiples of

$$\boldsymbol{v}_1 = \begin{bmatrix} 3 \\ 5 \end{bmatrix} \quad \text{and} \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Next step is write the given x_0 in terms of v_1 and v_2. there exist weights c_1 and c_2 such that

$$\boldsymbol{x}_0 = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 = [\boldsymbol{v}_1 \quad \boldsymbol{v}_2] \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \qquad (3)$$

In fact

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = [\boldsymbol{v}_1 \quad \boldsymbol{v}_2]^{-1}\boldsymbol{x}_0 = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}^{-1}\begin{bmatrix} 0.60 \\ 0.40 \end{bmatrix}$$
$$= \frac{1}{-8}\begin{bmatrix} -1 & -1 \\ -5 & 3 \end{bmatrix}\begin{bmatrix} 0.60 \\ 0.40 \end{bmatrix} = \begin{bmatrix} 0.125 \\ 0.225 \end{bmatrix} \qquad (4)$$

Because v_1 and v_2 in (3) are eigenvectors of A, with $Av_1=v_1$, and $Av_2=.92v_2$, we easily compute each x_k

$$\begin{aligned} \boldsymbol{x}_1 = A\boldsymbol{x}_0 &= c_1A\boldsymbol{v}_1 + c_2A\boldsymbol{v}_2 \\ &= c_1\boldsymbol{v}_1 + c_2(0.92)\boldsymbol{v}_2 \\ \boldsymbol{x}_2 = A\boldsymbol{x}_1 &= c_1A\boldsymbol{v}_1 + c_2(0.92)A\boldsymbol{v}_2 \\ &= c_1\boldsymbol{v}_1 + c_2(0.92)^2\boldsymbol{v}_2 \end{aligned}$$

And so on. In general

Using c_1 and c_2 from (4)

$$\boldsymbol{x}_k = c_1\boldsymbol{v}_1 + c_2(0.92)^k\boldsymbol{v}_2 \quad (k = 0,1,2,\cdots)$$

$$\boldsymbol{x}_k = 0.125\begin{bmatrix}3\\5\end{bmatrix} + 0.225(0.92)^k\begin{bmatrix}1\\-1\end{bmatrix} \quad (k = 0,1,2,\cdots) \qquad (5)$$

This explicit formula for x_k gives the solution of the difference equation $x_{k+1} = Ax_k$. As $k \to \infty$, $(.92)^k$ tends to zero and x_k tends to

$$(0.92)^k \to 0, \boldsymbol{x}_k \to \begin{bmatrix}0.375\\0.625\end{bmatrix} = 0.125\boldsymbol{v}_1.$$

Assignment

- Ex5.2: 18,23

§ 5.3 Diagonalization

- In many cases, we can get the eigenvalue and eigenvector of a matrix A of the form $A = PDP^{-1}$.

- In this section, we learn how to compute A^k quickly for large values of k. It is a fundamental idea in several applications of linear algebra.

The goal here is to develop a useful factorization $A = PDP^{-1}$, when A is $n \times n$. We can use this to compute A^k quickly for large k.

The matrix D is a *diagonal* matrix (i.e. entries off the main diagonal are all zeros).

$$\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & a_{nn} \end{bmatrix}$$

D^k is trivial to compute as the following example illustrates.

EXAMPLE: Let $D = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}$. Compute D^2 and D^3. In general, what is D^k, where k is a positive integer?

Solution:

$$D^2 = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 5^2 & 0 \\ 0 & 4^2 \end{bmatrix}$$

$$D^3 = D^2D = \begin{bmatrix} 5^2 & 0 \\ 0 & 4^2 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 5^3 & 0 \\ 0 & 4^3 \end{bmatrix}$$

and in general,

$$D^k = \begin{bmatrix} 5^k & 0 \\ 0 & 4^k \end{bmatrix}$$

EXAMPLE: Let $A = \begin{bmatrix} 6 & -1 \\ 2 & 3 \end{bmatrix}$. Find a formula for A^k given that $A = PDP^{-1}$ where $P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}$ and $P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$.

Solution:

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD(P^{-1}P)DP^{-1} = PDDP^{-1} = PD^2P^{-1}$$

Again,

$$A^3 = A^2A = (PD^2P^{-1})(PDP^{-1}) = PD^2(P^{-1}P)DP^{-1} = PD^3P^{-1}$$

In general,

$$A^k = PD^kP^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 5^k & 0 \\ 0 & 4^k \end{bmatrix}\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \cdot 5^k - 4^k & -5^k + 4^k \\ 2 \cdot 5^k - 2 \cdot 4^k & -5^k + 2 \cdot 4^k \end{bmatrix}.$$

A square matrix A is said to be **diagonalizable** if A is similar to a diagonal matrix, i.e. if $A = PDP^{-1}$ where P is invertible and D is a diagonal matrix.

§ 5.3 Diagonalization

When is A diagonalizable?

§ 5.3 Diagonalization

A square matrix A is diagonalizable if A is similar to a diagonal matrix, that is, if $A = PDP^{-1}$

THEOREM 5 The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

Proof ： (1) Now suppose A is diagonalizable, that is $A = PDP^{-1}$. Equivalently, we have AP = PD .

Let $P=[v_1 \ldots v_n]$ and D is a diagonal matrix with diagonal entries $\lambda_1, \ldots \lambda_n$. Then

$$AP = A[\boldsymbol{v}_1 \quad \boldsymbol{v}_2 \quad \cdots \quad \boldsymbol{v}_n] = [A\boldsymbol{v}_1 \quad A\boldsymbol{v}_2 \quad \cdots \quad A\boldsymbol{v}_n] \qquad (1)$$

and

$$PD = P\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} = [\lambda_1\boldsymbol{v}_1 \quad \lambda_2\boldsymbol{v}_2 \quad \cdots \quad \lambda_n\boldsymbol{v}_n] \qquad (2)$$

In this case,(1) and (2) imply that

$$[A\boldsymbol{v}_1 \quad A\boldsymbol{v}_2 \quad \cdots \quad A\boldsymbol{v}_n] = [\lambda_1\boldsymbol{v}_1 \quad \lambda_2\boldsymbol{v}_2 \quad \cdots \quad \lambda_n\boldsymbol{v}_n] \qquad (3)$$

That is

$$Av_1 = \lambda_1 v_1,\ Av_2 = \lambda_2 v_2, \cdots, Av_n = \lambda_n v_n$$

which means that $\lambda_1,\ldots,\lambda_n$ are eigenvalues and $v_1,\ldots,v_n$ are corresponding eigenvectors.

Since P is invertible, its columns $v_1,\ldots,v_n$ must be linearly independent. Therefore, A has n linearly independently eigenvectors.

(2) Suppose A has n linearly independent eigenvectors $v_1,\ldots,v_n$ and corresponding eigenvalues $\lambda_1,\ldots\lambda_n$.

Then let $P=[v_1\ldots v_n]$ and D is a diagonal matrix with diagonal entries $\lambda_1,\ldots\lambda_n$. Similarly, we have AP=PD.

Since the eigenvectors are linearly independent, then P is invertible, and AP = PD implies that $A = PDP^{-1}$.

EXAMPLE: Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

Step* 1. *Find the eigenvalues of A.

$$\det(A - \lambda I) = \det\begin{bmatrix} 2-\lambda & 0 & 0 \\ 1 & 2-\lambda & 1 \\ -1 & 0 & 1-\lambda \end{bmatrix} = (2-\lambda)^2(1-\lambda) = 0.$$

Eigenvalues of A: $\lambda = 1$ and $\lambda = 2$.

Step 2. Find three linearly independent eigenvectors of A.

By solving $(A - \lambda I)\mathbf{x} = \mathbf{0}$, for each value of λ, we obtain the following:

Basis for $\lambda = 1$: $\mathbf{v}_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$

Basis for $\lambda = 2$: $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $\mathbf{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

可验证三者线性无关

Step 3: Construct P from the vectors in step 2.

$$P = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Step 4: Construct D from the corresponding eigenvalues.

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Step 5: Check your work by verifying that AP = PD

$$AP = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -2 \\ -1 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

$$PD = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -2 \\ -1 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

§ 5.3 Diagonalization

EXAMPLE: Diagonalize the following matrix, if possible.

值得注意

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 0 & 2 & 2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Since this matrix is triangular, the eigenvalues are $\lambda = 2$ and $\lambda = 4$. By solving $(A - \lambda I)\mathbf{x} = \mathbf{0}$ for each eigenvalue, we would find the following:

Basis for $\lambda = 2$: $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Basis for $\lambda = 4$: $\mathbf{v}_2 = \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}$

Every eigenvector of A is a multiple of $\mathbf{v}_1$ or $\mathbf{v}_2$ which means there are not three linearly independent eigenvectors of A and by Theorem 5, A is not diagonalizable.

§ 5.3 Diagonalization

The Theorem 6 provides a sufficient condition for a matrix to be diagonalizable.

THEOREM 6 An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Proof： Let $v_1,\ldots,v_n$ be eigenvectors corresponding to the n distinct eigenvalues of a matrix A, then $\{v_1,\ldots,v_n\}$ is linearly independent, by Theorem 2 (Section 5.1) Hence A is diagonalizable, by Theorem 5.

§ 5.3 Diagonalization

EXAMPLE: Why is $A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 6 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ diagonalizable?

Solution: Since A has three eigenvalues ($\lambda_1 =$ 2, $\lambda_2 =$ 6, $\lambda_3 =$ 1) and since eigenvectors corresponding to distinct eigenvalues are linearly independent, A has three linearly independent eigenvectors and it is therefore diagonalizable.

If an n×n matrix A has the eigenvalues which are not distinct, we can use the Theorem 7 to build P that makes P automatically invertible.

THEOREM 7 Let A be an $n \times n$ matrix whose distinct eigenvalues are $\lambda_1, \ldots, \lambda_p$.

?

a. For $1 \le k \le p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of the eigenvalue λ_k.

b. The matrix A is diagonalizable if and only if the sum of the dimensions of the distinct eigenspaces equals n, and this happens if and only if the dimension of the eigenspace for each λ_k equals the multiplicity of λ_k.

c. If A is diagonalizable and β_k is a basis for the eigenspace corresponding to λ_k for each k, then the total collection of vectors in the sets $\beta_1, \ldots, \beta_p$ forms an eigenvector basis for $\mathbf{R}^n$.

证明复杂，略

§ 5.3 Diagonalization

EXAMPLE: Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 24 & -12 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Solution: Eigenvalues: -2 and 2 (each with multiplicity 2).

Solving $(A - \lambda I)\mathbf{x} = \mathbf{0}$ yields the following eigenspace basis sets.

Basis for $\lambda = -2$: $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -6 \\ 0 \end{bmatrix}$ $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 3 \\ 0 \end{bmatrix}$

Basis for $\lambda = 2$: $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ $\mathbf{v}_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is linearly independent

$\Rightarrow P = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{v}_4 \end{bmatrix}$ is invertible

$\Rightarrow A = PDP^{-1}$, where

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -6 & 3 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Assignment

- Ex5.3: 9,11,17,29

§ 5.4 Eigenvectors and Linear Transformations

$$A = PDP^{-1}$$

- **Objective**: We aim to understand the diagonalization of matrix *A from the* linear transformation view of point (从线性变换角度看矩阵的对角化).
- **The key point**(要掌握的主要知识点):

(1) **Matrix representation of linear transformations (线性变换的矩阵表示)。**

(2) 从线性变换的角度来看，$A = PDP^{-1}$意义为何？

- **1. The Matrix of a Linear Transformation**

Suppose

V ---- n-dimensional vector space,

W --- m-dimensional vector space,

T --- any linear transformation from V to W

β and C --- the bases for V and W.

$[x]_{\beta}$ --- the coordinate vector for any x in V (in R^n)

$[T(x)]_C$ --- the coordinate vector of its image (in R^m)

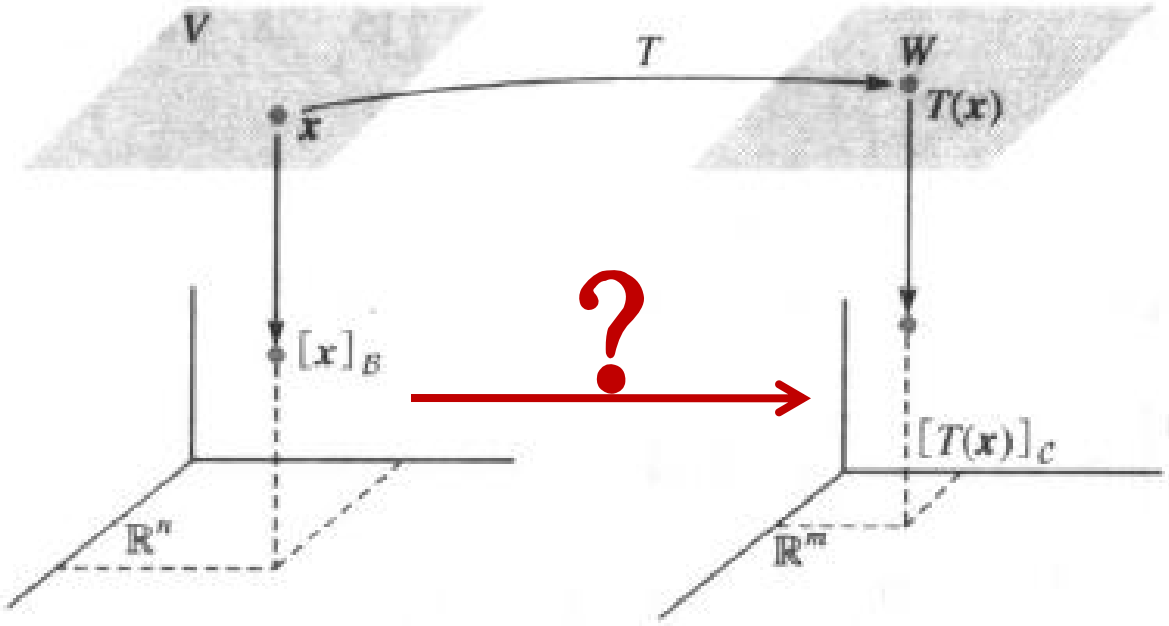


图 5-6　V 到 W 的线性变换

The connection between $[x]_B$ and $[T(x)]_C$

Let β={ $b_1,\ldots,b_n$ } be the basis for V.

If $x = r_1 b_1 + \cdots + r_n b_n$, then

$$[x]_B = \begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix}$$

Because T is linear，then

$$T(x) = T(r_1 b_1 + \cdots + r_n b_n) = r_1 T(b_1) + \cdots + r_n T(b_n)$$

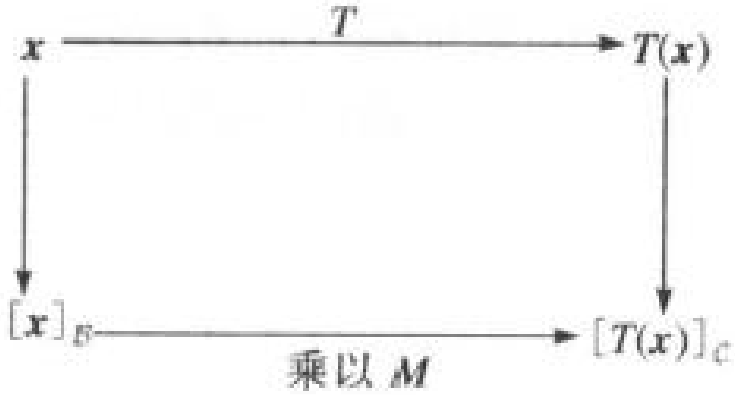
Using the basis C in W, we have

$$[T(x)]_C = r_1[T(b_1)]_C + \cdots + r_n[T(b_n)]_C$$

=> $[T(x)]_C = M[x]_B$

$$M = [[T(b_1)]_C \quad [T(b_2)]_C \quad \cdots \quad [T(b_n)]_C]$$

M: the matrix for T relative to the bases β and C



Example： Suppose $\beta = \{b_1, b_2\}$ is a basis for V and $C = \{c_1, c_2, c_3\}$ is a basis for W. Let T: V→W be a linear transformation defined by

$$T(\boldsymbol{b}_1) = 3\boldsymbol{c}_1 - 2\boldsymbol{c}_2 + 5\boldsymbol{c}_3,\ T(\boldsymbol{b}_2) = 4\boldsymbol{c}_1 + 7\boldsymbol{c}_2 - \boldsymbol{c}_3$$

Find the matrix M for T relative to β and C.

$$M = [[T(\boldsymbol{b}_1)]_C \quad [T(\boldsymbol{b}_2)]_C \quad \cdots \quad [T(\boldsymbol{b}_n)]_C]$$

Solution：

$$[T(\boldsymbol{b}_1)]_C = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}, \qquad [T(\boldsymbol{b}_2)]_C = \begin{bmatrix} 4 \\ 7 \\ -1 \end{bmatrix}$$

$$M = \begin{bmatrix} 3 & 4 \\ -2 & 7 \\ 5 & -1 \end{bmatrix}$$

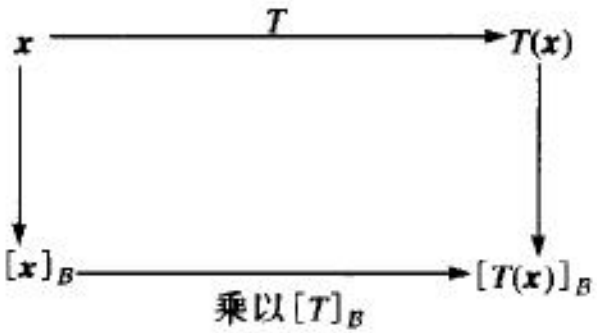
- **2. Linear Transformations from *V* into *V***

Let T: V→ V, let $\mathcal{B} = \{b_1, b_2 ... b_n\}$ be a basis of V.

The matrix for T relative to $\mathcal{B}$, or $\mathcal{B}$ -matrix for T is denoted by $[T]_{\mathcal{B}}$

Then the $\mathcal{B}$ -matrix for T: V→V is

$$[T]_{\mathcal{B}} = [T(b_1)_{\mathcal{B}}, T(b_1)_{\mathcal{B}} ... T(b_1)_{\mathcal{B}}]$$



$$[T(x)]_B = [T]_B [x]_B$$

Note that is a $[T]_{\mathcal{B}}$ is a square matrix. If the dimension of V is n, then $[T]_{\mathcal{B}}$ is $n \times n$matrix

- Example： The mapping T: $P_2 \to P_2$ is a linear transformation.

$$T(a_0 + a_1 t + a_2 t^2) = a_1 + 2a_2 t$$

a. Find the $\mathcal{B}$ -matrix for T, when $\mathcal{B}$ is the basis $\{1, t, t^2\}$.

b. Verify that for each p in P_2

$$[T(p)]_{\mathcal{B}} = [T]_{\mathcal{B}} [p]_{\mathcal{B}}.$$

Solution： a. Compute the images of the basis vectors:

$$T(1) = 0 \qquad \text{零多项式}$$

$$T(t) = 1 \qquad \text{值恒为 1 的多项式}$$

$$T(t^2) = 2t$$

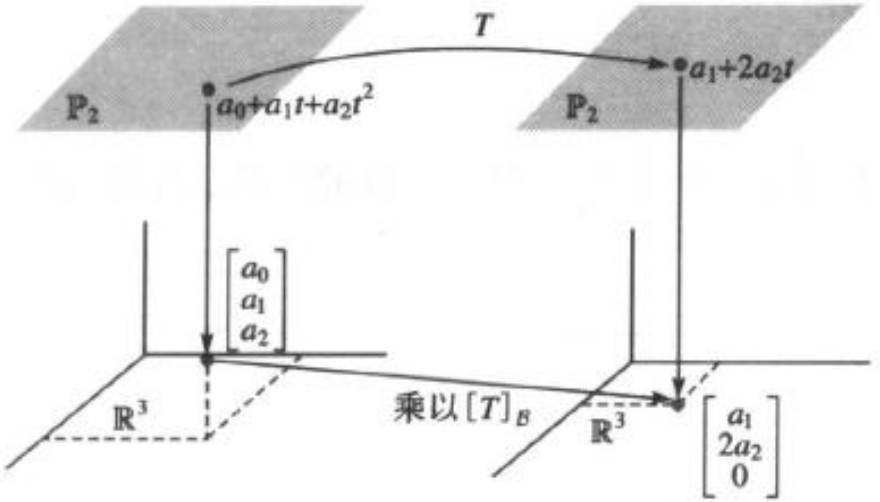
$$[T(1)]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad [T(t)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad [T(t^2)]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

$$[T]_{\mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

- b. For a general polynomial $p(t) = a_0 + a_1 t + a_2 t^2$, we have

$$[T(\boldsymbol{p})]_B = [a_1 + 2a_2 t]_B = \begin{bmatrix} a_1 \\ 2a_2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = [T]_B [\boldsymbol{p}]_B$$



- **3. Linear Transformations on R^n**

Theorem 8 Diagonal Matrix Representation

Suppose $A = PDP^{-1}$, where D is a diagonal $n \times n$ matrix.

Then D is the $\mathcal{B}$-matrix for the transformation $T: x \rightarrow Ax$,

where $\mathcal{B}$ is the basis for R^n formed from the columns of P.

A、D是同一个线性变换在不同基下的矩阵表示

思路：$[T]_{\mathcal{B}} = [T(b_1)_{\mathcal{B}}, T(b_1)_{\mathcal{B}} \ldots T(b_1)_{\mathcal{B}}]$=⋯ … = D

Proof：Denote the columns of P by $b_1, \ldots, b_n$, so that

$$\mathcal{B} = \{\boldsymbol{b}_1, \cdots, \boldsymbol{b}_n\}, P = [\boldsymbol{b}_1 \cdots \boldsymbol{b}_n].$$

P is the change-of-coordinates matrix $P_{\mathcal{B}}$, where

$$P[\boldsymbol{x}]_{\mathcal{B}} = \boldsymbol{x} \text{ 和 } [\boldsymbol{x}]_{\mathcal{B}} = P^{-1}\boldsymbol{x}$$

Then, we have

$$\begin{aligned}
[T]_{\mathcal{B}} &= [[T(\boldsymbol{b}_1)]_{\mathcal{B}} \cdots [T(\boldsymbol{b}_n)]_{\mathcal{B}}] && \text{由 } [T]_{\mathcal{B}} \text{ 的定义} \\
&= [[A\boldsymbol{b}_1]_{\mathcal{B}} \cdots [A\boldsymbol{b}_n]_{\mathcal{B}}] && \text{由 } T(\boldsymbol{x}) = A\boldsymbol{x} \\
&= [P^{-1}A\boldsymbol{b}_1 \cdots P^{-1}A\boldsymbol{b}_n] && \text{坐标变换} \\
&= P^{-1}A[\boldsymbol{b}_1 \cdots \boldsymbol{b}_n] && \text{矩阵乘法} \\
&= P^{-1}AP
\end{aligned}$$

由 $A = PDP^{-1}$，我们有 $[T]_{\mathcal{B}} = P^{-1}AP = D$.

Example: Define T: $R^2 \rightarrow R^2$ by $T(x) = Ax$, where $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$.

Find a basis $\mathcal{B}$ for R^2 with the property that the $\mathcal{B}$ -matrix for T is a diagonal matrix.

Diagonalize A（ 对A进行对角化分解 ）

Similarity of Matrix Representations

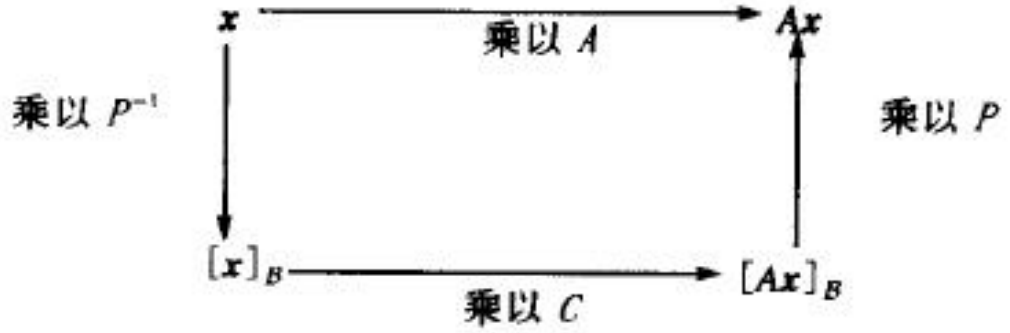
从线性变换的角度来看，相似变换$A = PCP^{-1}$表示什么意思？

由刚才的Th8的证明，可以得到以下一般化的结论：

If A and C are similar, that is $A = PCP^{-1}$, then

C is the $\mathcal{B}$ -matrix for the transformation T:x→Ax, where the basis $\mathcal{B}$ is formed from the columns of P.

The factorization $A = PCP^{-1}$is shown below.



- Conversely, if T: $R^n \rightarrow Rn$ is defined by T(x) = Ax, and if $\mathcal{B}$ is any basis for R^n, then the $\mathcal{B}$ -matrix for T is similar to A.
- In fact , if P is the matrix whose columns come from the vectors in $\mathcal{B}$, then .

$$[T]_B = P^{-1}AP$$

若矩阵A，B相似，则他们是R^n上同一线性变换在不同基下的矩阵表示；反之，也成立。

Assignment

- Ex5.4:1,7,15

§5.5 Complex Eigenvalues
（ 简单了解 ）

- If the characteristic equation of an n×n matrix involves a polynomial of degree n, the equation always has exactly n roots, counting multiplicities. (复数根计算在内)

- Complex eigenvalues is essential to uncover "hidden" information about certain matrices.

（ 实矩阵可以有复特征值 ）

Some notations

- C^n -- n-dimension complex space (n维复空间)
- λ -- complex scalar

  (complex) eigenvalue （复）特征值
- x -- nonzero vector x in C^n

  (complex) eigenvector （复）特征向量
- λ satisfies $\det(A-\lambda I) = 0$ if and only if $Ax = \lambda x$.

- Example： Let $A=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$ The characteristic equation of A is

$$\lambda^2 + 1 = 0.$$

The only roots are complex: $\lambda = i$ and $\lambda = -i$.

$$\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1\\ -i\end{bmatrix}=\begin{bmatrix}i\\ 1\end{bmatrix}=i\begin{bmatrix}1\\ -i\end{bmatrix}$$
$$\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1\\ i\end{bmatrix}=\begin{bmatrix}-i\\ 1\end{bmatrix}=-i\begin{bmatrix}1\\ i\end{bmatrix}$$

Thus i and -i are eigenvalues, with $\begin{bmatrix}1\\ -i\end{bmatrix}$ and $\begin{bmatrix}1\\ i\end{bmatrix}$

不讲

5.6，5.7关于特征值、特征向量的应用

5.8特征值的估算（ 数值计算 ）